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Hyperconvexity and domains of discontinuity for surface groups in $SL_3\mathbb{C}$

BY

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THESIS [PhD candidates may use DISSERTATION here instead]

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Accessibility Statement

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Dedication

For my grandmother Martha, who also saw the joy in geometry.

Acknowledgements

Emily (integrity, knowledge, high standards)
Sam (love, support, meals, no pressure)
Family (you don't know why I wanted to do this, but that's ok)
Friends (kept me company, kept me happy)
Other math teachers and mentors
Gertrude (tiniest companion)

Preface

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List of Abbreviations

- $\mathbb{R}$ The real numbers
- $\mathbb{C}$ The complex numbers

Summary

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CHAPTER 1

Introduction

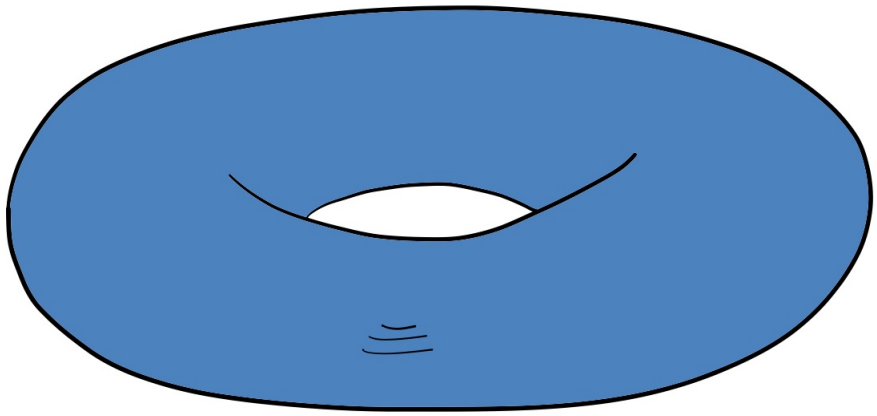

FIGURE 1. This is a torus

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & A' \arrow[r, "f'"] \arrow[d, "a"'] & B' \arrow[r, "g'"] \arrow[d, "b"'] & C' \arrow[r] \arrow[d, "c"'] & 0 \\
0 \arrow[r] & A \arrow[r, "f"] \arrow[d, "a''"'] & B \arrow[r, "g"] \arrow[d, "b''"'] & C \arrow[r] \arrow[d, "c''"'] & 0 \\
0 \arrow[r] & A'' \arrow[r, "f''"] \arrow[d] & B'' \arrow[r, "g''"] \arrow[d] & C'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

FIGURE 2. The Snake Lemma

This is the introduction chapter. We cite some classic works [**1, 2**].

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Morbi tincidunt nisi sed neque volutpat, et imperdiet metus congue. Donec interdum, neque at varius vehicula, elit odio tempus sapien, sed finibus lorem felis nec turpis. Aliquam erat volutpat. Mauris vehicula imperdiet erat, non aliquet magna commodo id.

Sed porttitor urna ac ipsum rhoncus, vel dictum risus commodo. Integer ultrices dui sit amet pulvinar tempor. Phasellus nec tellus eu turpis sodales accumsan. Proin ultr

THEOREM 1.1. *This is a theorem*

We reference Theorem 1.1.

University of Illinois Chicago.

1. Motivation

1.1. Historical context. A brief overview of how the problem developed.

$$\int_0^1 f(x)dx = 2 \tag{1}$$

How to solve (1)

1.2. Open questions. Some questions remain open for future work.
Note: I have not tested the accessibility of this table.

Monkeys	Lions
100	200

Table 1. Example Table

CHAPTER 2

Background

This chapter gives necessary background.

1. Group theory

A group is a set G with a binary operation satisfying closure, associativity, identity, and inverses.

THEOREM 2.1. *Every finite subgroup of the multiplicative group of a field is cyclic.*

PROOF. This is a standard result from algebra. $\square$

CHAPTER 3

Main Result

Let Γ the fundamental group of a closed hyperbolic surface. Let $\rho, \rho' : \pi_1\Sigma \to SL_3\mathbb{C}$ Anosov. We use $\xi, \xi' : \partial\Gamma \to \mathcal{F}^3$ to mean the continuous, transverse, ρ- or ρ'- equivariant, dynamics-preserving boundary maps. We use $\Lambda, \Lambda' \subset \mathcal{F}^3$ to mean the limit curves, which are the images of ξ, ξ', respectively. We use $Th(\Lambda)$ to refer to the thickened limit curve. We use $Th(x)$ to refer to the thickening of a particular flag $x \in \Lambda$.

We call ρ, ρ' *topologically conjugate* if there exists a homeomorphism $F : \mathcal{F}^3 \to \mathcal{F}^3$ such that for all $g \in \Gamma$ and all $x \in \mathcal{F}^3$, then $F(\rho(g) \cdot x) = \rho'(g) \cdot F(x)$.

THEOREM 3.1. *Hyperconvexity is closed under topological conjugacy. In other words, if ρ and ρ' are topologically conjugate, then ρ is hyperconvex if and only if ρ' is hyperconvex.*

We assume $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between ρ and ρ'. Clearly $F(\Lambda) = \Lambda'$. This mapping is pointwise, i.e. for all $\alpha \in \partial\Gamma$, then $F(\xi(\alpha)) = \xi'(\alpha)$. So for $x \in \Lambda$, we can use the notation $x' := F(x) \in \Lambda'$.

For any pair of transverse flags $x, y \in \mathcal{F}^3$, consider the irreducible 0-dim Richardson varieties $R^0_{x,y} := \{x, x_y, x^y, y_x, y^x, y\}$. If x and y are the attracting and repelling flags of $diag(2, 1, 0.5)$ then $R^0(x, y)$ consists of:

- $x = \{x\} \cap \alpha_{121}(y) = (e_1, e_1 + e_2)$
- $x_y = \alpha_1(x) \cap \alpha_{12}(y) = (e_1, e_1 + e_3)$
- $x^y = \alpha_2(x) \cap \alpha_{21}(y) = (e_2, e_1 + e_2)$
- $y_x = \alpha_{12}(x) \cap \alpha_1(y) = (e_3, e_1 + e_3)$
- $y^x = \alpha_{21}(x) \cap \alpha_2(y) = (e_2, e_2 + e_3)$
- $y = \alpha_{121}(x) \cap \{y\} = (e_3, e_2 + e_3)$.

LEMMA 3.1.1. *As a set, $F(R^0_{x,y}) = R^0_{x',y'}$.*

However, F may not pointwise preserve the "type" of Richardson variety. For all $x, y \in Lambda^{(2)}$,

- *if F* matches *Richardson varieties, then $F(x_y) = x'_{y'}$, $F(x^y) = x'^{y'}$, $F(y_x) = y'_{x'}$, and $F(y^x) = y'^{x'}$.*
- *if F* switches *Richardson varieties, then $F(x_y) = x'^{y'}$, $F(x^y) = x'_{y'}$, $F(y_x) = y'^{x'}$, and $F(y^x) = y'_{x'}$.*

As a consequence of the argument in Lemma 3.1.1, we also have $F(Th(x)) = Th(x')$ for all $x \in \Lambda$.

LEMMA 3.1.2. *The representation ρ is hyperconvex if and only if for all $x, y, z \in \Lambda^{(3)}$, then $R^0_{x,y} \cap R^0_{x,z} = \{x\}$.*

PROOF. (Theorem 3.1) Suppose $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between ρ and ρ'.

We begin with the forward direction. Assume that ρ is hyperconvex. Let $x, y, z \in \Lambda^{(3)}$. By Lemma 3.1.2, then $x_y \neq x_z$ and $x^y \neq x^z$. By Lemma **??**, then $\{F(x_y), F(x^y), F(x_z), F(x^z)\} = \{x'_{y'}, x'^{y'}, x'_{z'}, x'^{z'}\}$ are distinct. Since x, y, z were general, then this holds for all triples in Λ and ρ' is hyperconvex.

The other direction proceeds similarly. Assume that ρ is *not* hyperconvex. By Lemma 3.1.2, there exist $x, y, z \in \Lambda^{(3)}$ such that $x_y = x_z$ or $x^y = x^z$. By Lemma 3.1.1, then F either matches or switches Richardson varieties. So one of the following is true: $x'_{y'} = x'_{z'}$ or $x'^{y'} = x'^{z'}$. Therefore ρ' is *not* hyperconvex. $\square$

Proof. (Lemma 3.1.1) Let's describe the dynamics of $\rho(g) \cdot \mathcal{F}^3$, where $g \in \Gamma$ infinite order. Since ρ is Anosov, then we know $\rho(g)$ is strongly loxodromic, i.e. its eigenvalues have distinct magnitude. That means it has an attracting and repelling fixed boundary points $g^+, g^- \in \partial\Gamma$. Let $x = \xi(g^+), y = \xi(g^-)$.

We define the "uphill" $U_g(x) := \{f : \rho(g^n) \cdot f \to x\}$ and "downhill" $D_g(y) := \{f : \rho(g^{-n}) \cdot f \to y\}$. Most flags are in the open and dense set $U_g(x) \cap D_g(y) \subset \mathcal{F}^3$. However, there is an exceptional locus $E_g = \mathcal{F}^3 \setminus (U_g(x) \cap D_g(y))$, i.e. the attracting or repelling flags for E_g are other fixed points under $\rho(g)$. The fixed points of $\rho(g) \cdot \mathcal{F}^3$ are exactly $R^0_{x,y}$.

These happen to be the irreducible 1-dim Richardson varieties of x, y:

- $\alpha_1(x) = \alpha_1(x) \cap \alpha_{121}(y) = \overline{U(x) \cap D(x_y)}$
- $\alpha_2(x) = \alpha_2(x) \cap \alpha_{121}(y) = \overline{U(x) \cap D(x^y)}$
- $span(x_y, y_x) = \alpha_{12}(x) \cap \alpha_{12}(y) = \overline{U(x_y) \cap D(y_x)}$
- $span(x^y, y^x) = \alpha_{21}(x) \cap \alpha_{21}(y) = \overline{U(x^y) \cap D(y^x)}$
- $\alpha_1(y) = \alpha_{121}(x) \cap \alpha_1(y) = \overline{U(y_x) \cap D(y)}$
- $\alpha_2(y) = \alpha_{121}(x) \cap \alpha_2(y) = \overline{U(y^x) \cap D(y)}$

Claim: $F(Th(x)) = Th(x')$
Claim: $F(R^0_{x,y}) = R^0_{x',y'}$.
Claim: $F(R)$
Claim: $F(R^0_{\rho(h)\cdot x, \rho(h)\cdot y}) = R^0_{\rho'(h)\cdot x', \rho'(h)\cdot y'}$.
Claim: Pairs of the form (g^+, g^-) are dense in $\partial\Gamma^2$

□

Proof. (Lemma 3.1.2) Suppose ρ is Anosov. Let $x = (x_1, x_2), y = (y_1, y_2), z = (z_1, z_2) \in \Lambda^{(3)}$. Since x, y, z are pairwise transverse, then clearly $y_x, y^x, z_x, z^x \notin R^0_{x,y} \cap R^0_{x,z}$. It suffices to check the pairs $x_y, x_z \in \alpha_1(x)$ and $x^y, x^z \in \alpha_2(x)$.

By definition, $x_y = (x_1, x_1 + y_1)$ and $x_z = (x_1, x_1 + z_1)$, which are distinct if and only if $x_1 + y_1 \neq x_1 | z_1$, or equivalently $z_1 \not\subset x_1 + y_1$.

Similarly, $x^y = (x_2 \cap y_2, x_2)$ and $x^z = (x_2 \cap z_2, x_2)$, which are distinct if and only if $x_2 \cap y_2 \neq x_2 \cap z_2$, or equivalently $x_2 \cap y_2 \not\subset z_2$.

So clearly $R^0_{x,y} \cap R^0_{x,z} = \{x\}$ for all $x, y, z \in \Lambda^{(3)}$ is equivalent to ρ hyperconvex. □

CHAPTER 4

Another result

1. Second main result

Another significant theorem.

CHAPTER 5

Vitae

For UIC this can be a full CV or a short version (e.g. half a page) with previoius degrees, work experience and publications.

APPENDIX A

Technical Lemmas

Here we collect some supporting lemmas.

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